

Week 3. Class 6. Biocompatible materials in Engineering Design

Biocompatibility n. <i>/ˌbaɪəʊkəmˌpætəˈbɪləti/</i>	How well a material performs in the body without causing unacceptable harm.	Biocompatibility is the first requirement for any implant material.
Adverse reactions n. pl. <i>/ˈædvɜːs riˈækʃənz/</i>	Harmful or unwanted responses of the body to a material or device.	Adverse reactions may occur even when the device works mechanically.
Inflammation n. <i>/ˌɪnfləˈmeɪʃən/</i>	A biological reaction (often swelling, redness, pain) that can occur when the immune system is activated.	Inflammation around an implant can damage surrounding tissue.
Rejection n. <i>/rɪˈdʒekʃən/</i>	A strong immune response that attacks an implanted material or device.	In severe cases, rejection can lead to device failure.
Immunogenicity n. <i>/ˌɪmjʊːnəˈdʒenɪsɪti/</i>	The likelihood that a material will trigger an immune response.	Engineers test immunogenicity to reduce long-term risk.
Bioactive coating n. <i>/ˌbaɪəʊˈæktɪv ˈkoʊtɪŋ/</i>	A surface layer that is designed to actively improve biological response (e.g., support integration).	A bioactive coating can help tissue integrate with an implant.
Leach v. <i>/liːtʃ/</i>	To release substances from a material into surrounding fluids/tissues over time.	Some polymers can leach chemicals into body fluids.
Leaching n. <i>/ˈliːtʃɪŋ/</i>	The process of releasing substances from a material into surrounding fluids/tissues over time.	Leaching can increase toxicity risk if harmful compounds are released.
Contamination n. <i>/kənˌtæməˈneɪʃən/</i>	Unwanted substances introduced during manufacturing/processing that can create safety risks.	Strict protocols reduce contamination during production.
Microbial colonization n. <i>/maɪˈkrəʊbiəl ˌkɑːlənəˈzeɪʃən/</i>	When microorganisms like bacteria attach to and grow on a material's surface.	Microbial colonization increases infection risk after surgery.
Antimicrobial coatings n. pl. <i>/ˌæntaɪmaɪˈkrəʊbiəl ˈkoʊtɪŋz/</i>	Surface layers that kill bacteria or prevent bacterial growth.	Antimicrobial coatings are used to lower infection rates.
Hydrophilicity n. <i>/ˌhaɪdrəʊfɪˈlɪsɪti/</i>	Water-attracting surface behavior that often influences how cells and bacteria interact with a material.	Hydrophilicity can affect protein adsorption on a surface.

Scaffold n. /'skæfəld/	A temporary structure that supports cells and guides tissue growth in tissue engineering.	A scaffold provides support while new tissue forms.
Porosity n. /pɔ:'rɑ:siti/	The presence of pores/voids in a material, which can support tissue ingrowth or drug release.	Porosity must be tuned to balance strength and transport.
Autoclaving n. /'ɔ:toʊ ,kleɪvɪŋ/	Steam sterilization at high temperature and pressure.	Autoclaving is effective but may damage some polymers.
Ethylene oxide (EtO) n. /'eθəli:n 'ɑ:ksaɪd/; /i: ti: 'oʊ/	Gas sterilization method often used for heat-sensitive materials.	Ethylene oxide is used when heat would deform the device.
Total hip replacement n. /,toʊtəl hɪp rɪ 'pleɪsmənt/	A common medical implant that replaces the hip joint to restore function and reduce pain.	A total hip replacement can restore mobility after joint damage.
Regenerative biomaterials n. pl. /rɪ 'dʒenərətɪv ,baɪoʊsmə 'tɪriəlz/	Biomaterials engineered to help the body regenerate tissue, ideally back to its original form.	Regenerative biomaterials support healing rather than only replacement.
Biodegrade v. /,baɪoʊdɪ 'greɪd/	To break down gradually inside the body over time.	Some scaffolds biodegrade after they guide tissue growth.
Biodegradation n. /,baɪoʊ ,deɪgrə'deɪʃən/	The gradual breakdown of a material inside the body over time.	Biodegradation rate should match the pace of tissue repair.
Injected adj. /ɪn 'dʒektɪd/	Delivered into the body using a needle/syringe.	The nanoparticles are injected to reach the target tissue.
Blood–brain barrier n. /,blʌd 'breɪn 'bæriər/	A protective barrier that blocks many substances from entering the brain from the bloodstream.	Many drugs cannot cross the blood–brain barrier .
Bioglass n. /'baɪoʊsglæs/	A glass-like biomaterial that can dissolve in the body, bond to bone, and stimulate new bone growth.	Bioglass can stimulate bone regeneration through released ions.
Dissolve v. /dɪ'zɑ:lɪv/	To gradually break apart and mix into a liquid in the body; the material can dissolve over time.	Over time, the material may dissolve in body fluid.
	To attach strongly to bone tissue, forming a stable interface.	Some coatings help an implant bond to bone more effectively.

Bond to bone v. phrase /ba:nd tə boʊn/		
Sol–gel process n. /ˌsɑ:l ˈdʒel ˈpra:sɪs/	A chemistry-based method of making glass-like materials by building a network through chemical assembly.	The sol–gel process can produce bioactive glass structures.
Brittle adj. /ˈbrɪtəl/	Hard but easy to crack or break.	Glass can be brittle under repeated stress.
Synergy n. /ˈsɪnərdʒi/	A combined effect where two materials together perform better than each one alone.	A polymer–glass composite may show synergy in properties.
Surface chemistry n. /ˈsɜːfɪs ˈkɛmɪstri/	The chemical properties of a material’s surface that affect how cells interact with it.	Surface chemistry influences cell attachment and growth.
Biosensing n. /ˈbaɪoʊ ˌsensɪŋ/	Detecting biological signals (e.g., markers of disease or pathogens) using devices/materials.	Biosensing can help detect biomarkers in body fluids.
Biosensors n. pl. /ˈbaɪoʊ ˌsensərz/	Devices/materials used to detect biological signals (e.g., markers of disease or pathogens).	Biosensors are used to identify disease markers quickly.
Surgical intervention n. /ˈsɜːdʒɪkəl ˌɪntər ˈvenʃən/	A surgery required when the body cannot heal/regenerate a defect on its own.	Early treatment can sometimes reduce the need for surgical intervention.